

NeuroViG - Integrating Event Cameras for Resource-Efficient Video Grounding

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Abstract

Spatio-Temporal Video Grounding (STVG) – the task of identifying the target object in the field-of-view, that the language instruction refers to – is a fundamental vision-language task. Current STVG approaches typically utilize feeds from an RGB camera that is assumed to be always-on and process the video frames using complex neural network pipelines. As a result, they often impose prohibitive system overheads (energy, latency) on pervasive devices. To address this, we propose NeuroViG with two key innovations: (a) leveraging on event streams from a low-power neuromorphic event camera sensor to perform selective triggering of the more energy-hungry RGB camera for STVG, and (b) augmenting the STVG model with a lightweight Adaptive Frame Selector (AFS) that bypasses complex transformer-based operations for a majority of video frames, thereby enabling its execution on a pervasive Jetson AGX device. We have also introduced modifications to the neural network processing pipeline such that the system can offer tunable tradeoffs between accuracy and energy/latency. Our proposed NeuroViG system allows us to reduce the STVG energy overhead and latency by $\sim 4x$ and $\sim 3.8x$, respectively, for less than 1% loss in accuracy.

1. Introduction

Spatio-Temporal Video Grounding (STVG) is a compound vision-language task that combines both spatial and temporal grounding of a natural language query over a video. In STVG, a given video feed and a natural language query are taken as input to identify (a) the target object/region-of-interest, and (b) the start and end of the temporal segment within which this object is visible. Thus, the output of this multi-modal task is often visualized as a spatio-temporal tube extracted from the video by identifying the bounding-box of the target object over several consecutive frames of the relevant video segment.

STVG requires both temporal and spatial reasoning over multiple video frames. Implementing STVG on resource-constrained devices is thus very challenging due to high overheads involved in both sensing and processing. Single-stage transformer based models such as TubeDETR [24] and STCAT [9] dominate the performance charts in this challenging vision-language task. However, these state-of-the-art STVG models are very computationally intensive as they require cross-modal attention mechanisms to be repeated across several frames of a video. Typically, they load entire videos and execute the compute-heavy transformer encoders on *every single frame* to retain the resulting intermediate feature maps. Subsequently, the intermediate attention-based features obtained from the decoder are used by prediction heads to generate the following three outputs for every single video frame— (a) a bounding box, (b) a probability that this frame is the start-frame of a tube and (c) the probability that this frame is the end-frame of a tube. Our evaluation of TubeDETR [24] and STCAT [9] (see Figure 7) reveals that processing latency steadily increases with the video length. It is thus crucial to reduce the computational latency and energy of STVG models, especially for practical use-cases that (a) involve long untrimmed videos (e.g., security footage or instructional videos), or (b) weave such grounding into interactive applications (e.g., having service bots respond to human assistance requests). Besides the processing overhead, traditional STVG models also require continuous input from an RGB camera, which unnecessarily increases the sensing energy overhead. Thus, it is necessary to reduce *both* processing and sensing overheads.

In this paper, we propose *NeuroViG* - a system that can support the on-device execution of an STVG task on a pervasive device, with significantly lower processing and sensing overheads. Our key insight is that *it is often sufficient to execute the STVG pipeline on a small set of candidate moments/frames within a given video (e.g., when we see a frame that contains relevant objects or gestures men-*

tioned in the accompanying text). Therefore, instead of continuously capturing images with the RGB camera, we use an event camera such as the Dynamic Vision Sensor (DVS) [12] to trigger the RGB camera only when required. In contrast to RGB cameras which rely on successive image frames to capture the dynamics of a scene, event cameras can directly capture scene dynamics. Hence, using an event camera for grounding enables the direct extraction of motion-specific representations from the sensor’s output, eliminating the need for further processing of RGB features. They also have advantages such as – a) low latency (in microseconds), and b) low power consumption (0.31W in comparison to 1.1W in traditional cameras). However, completely replacing the traditional RGB camera with an event camera is not feasible, as they may not capture static components of a scene or attributes such as color. Furthermore, current computer vision pipelines are tailored for sense-making using traditional RGB camera input and not an event-camera stream. Hence, we opt for an opportunistic triggering based mechanism that harnesses the advantages of both an event camera and the RGB-based SOTA models for STVG. In this approach, the RGB camera helps capture static scene components and *key moments* in a video, whereas an event camera provides a low power approximation of the redundant frames.

To support selective triggering of the RGB camera, we have developed a lightweight early estimator named **Adaptive Frame Selector (AFS)** which takes the event-camera stream and the textual query as an input to predict the key moments known as *anchor-frames* within the video. The RGB camera is triggered upon detection of such anchor frames. Further, we propose a lightweight STVG model, called *NeuroViG*, which combines AFS along with the transformer encoder-decoder used by SOTA models for STVG. *NeuroViG* saves significant processing overheads by executing the complex transformer encoder-decoder pipelines exclusively on the anchor frames predicted by AFS. For the non-anchor frames, AFS provides a feature representation (called delta features) of the visual changes between the current frame and the last identified anchor frame. These delta features can then be used to generate an approximation of the attention-based features for each non-anchor frame. The following are our main contributions:

- **NeuroViG** : We believe that *NeuroViG* is the first system to execute the STVG task on a pervasive device (Jetson AGX device), running at 8.2 FPS while consuming a total energy of 1.18J per frame. This makes *NeuroViG* $\sim 3.82\times$ faster and $4.08\times$ energy efficient than STCAT, while maintaining similar accuracy.
- **Event camera based Adaptive Frame Selector**: We have developed a novel Adaptive Frame Selector to identify critical moments within the video that are pivotal

for accurately grounding the natural language instruction. Our modified loss function for the AFS enables us to choose between various levels of aggressive or conservative frame-skipping behaviour in the STVG model to exploit the trade-off between accuracy and overall energy consumption (sensing + processing).

2. Related Work

Grounding is a foundational task in vision-language understanding, wherein the agent’s goal is to identify elements within a video or image that correspond to a given linguistic expression. Early approaches in this domain [14, 16, 26] sought to leverage the success of pre-trained deep learning models for object detection and natural language processing to perform object grounding in images. Recent advancements in this research area have focused on enhancing efficiency [25, 30] and extending the task to encompass additional modalities, such as incorporating pointing gestures alongside verbal instructions [6, 22]. Zhang et al. [27] introduced the VidSTG dataset and the STGRN model [27], which employs a two-stage Deep Neural Network (DNN) pipeline to spatially and temporally locate objects. This method relies on region or tubelet proposals generated in the first stage, which are then used in the subsequent stage to find the best match with the provided linguistic expression. Transformer-based models also gained traction, including STVGBert [18], an extension of ViLBERT that preserves spatial information. STGVT [19] focuses on human-centric STVG with the HC-STVG dataset.

More recently, single-stage STVG approaches have become prominent, offering superior performance compared to earlier two-stage methods. TubeDETR [24], for example, extends MDETR [10] by incorporating temporal localization losses, slow-fast encoding, and space-time decoding. Another recent single-stage model for STVG is STCAT [9], which employs an MDETR-based framework and achieves state-of-the-art performance on both VidSTG and HC-STVG datasets. In contrast to TubeDETR, STCAT takes into account the global video context and maintains a template mechanism in the query-guided decoder to explicitly model consistency.

With the introduction of neuromorphic event cameras, many vision-based applications that traditionally relied on RGB cameras are beginning to adopt event camera streams as inputs. Event cameras offer several advantages over frame-based counterparts, including exceptionally fine temporal resolution (on the order of μsecs), a wide dynamic range, and low power consumption. Due to their asynchronous nature, numerous efforts have been made to process event-based input using spiking neural networks (SNNs) [20]. Prior to integrating event streams into processing pipelines for various vision-based applications, such raw event data can be transformed into: a) Frame-

based representations [32], b) Time-surface representations [17], or c) Voxel-based methods [15]. Recent advances in leveraging event cameras for sense-making have also extended to the fusion of RGB and event camera streams. For instance, Zhou et al. [31] proposed a frame-based multi-modal DNN pipeline that simultaneously processes both a conventional RGB camera stream and an event stream for tasks like moving object detection.

In line with these developments, our *NeuroViG* framework is built upon a frame-based representation for processing event streams and represents the first approach to present a DNN pipeline that integrates three distinct sensing modalities: RGB camera data, event camera data, and natural language text instructions. Remarkably, our *NeuroViG* system stands out as the first to use event camera streams for STVG instead of just using traditional RGB inputs. This innovative approach not only reduces sensing and processing energy overhead but also significantly improves latency.

3. Baseline

We consider several state-of-the-art STVG models as baselines. All of the existing models proposed for STVG use inputs from a traditional RGB camera, along with language instruction. We identified STCAT [9] and TubeDETR [24] as the current state-of-the-art STVG models. Both of them use single-stage architectures with a) ROBERTa [13] model for language encoding, b) ResNet-101 model as visual feature backbone, c) Transformer encoder for calculating multi-modal video-text features, d) Transformer decoder for modelling temporal dependencies and e) 2 prediction heads for calculating spatio-temporal bounding box tubes. Despite their single-stage architectures, these networks are generally computationally intensive due to the following reasons.

1. These approaches consist of a complex ResNet-101 [7] visual backbone, as well as multiple transformer encoder-decoder blocks for modelling multi-modal (video + text) features and temporal dependencies.
2. These networks demand all the frames of the input video (partially or fully) to pass through a resource-intensive pipeline involving multiple transformer blocks. TubeDETR [24] identified that this is very computationally intensive and used a fast-slow two-stream approach in their encoder that samples the video uniformly such that only one out of every k frames (k is an ad hoc constant) passes through the transformer encoder (slow multi-modal stream) to extract the attention-based features. However, TubeDETR still demands all the frames to pass through a complex RESNET backbone and a transformer decoder. On the other hand, STCAT, which achieves higher accuracy, demands all of the frames

to pass through its complete pipeline including the encoder and decoder. Such computation imposes an average latency of 390msec per image and memory of 2.1GB for TubeDETR, and 450msec latency per image and memory of 2.8GB STCAT, for VidSTG dataset on an NVIDIA Jetson AGX device [3].

3. None of the existing work has looked at reducing the sensing overhead. Both STCAT and TubeDETR (and all of the existing work), require the camera to be always turned on. This will not be suitable, particularly in a pervasive setting where the sensing is performed by a battery-operated wearable or embedded device.

To overcome these limitations, we looked into suitable approaches for jointly reducing the computational and sensing load involved in spatiotemporal video grounding. While approaches like model pruning and skipping [21] have been studied for unimodal tasks, they are less suited for DNNs used in multi-modal tasks such as STVG. Some previous works such as SoftSkip [23] and COSM2IC [22], have highlighted this problem and have come up with specialized approaches for multi-modal tasks such as referring expression comprehension. Even so, such optimization has not been attempted for transformer-based architectures that are common in STVG or for reducing sensing overhead. However, we observed that: (a) the main computational load arises from the need for all the frames to pass through the whole pipeline, and (b) the main sensing overhead comes from having to keep a traditional RGB camera always turned on. We develop our *NeuroViG* to address these drawbacks.

4. *NeuroViG* System

We now describe the design of our proposed *NeuroViG* system. We propose a hybrid sensing system, consisting of both an event camera and an intermittently triggered RGB camera. Our key idea is to use the RGB sensing stream for some key moments in the video which we termed as *anchor frames*. These anchor frames may represent frames which either will not be captured well through an event camera (static scenes which may generate few events) or where relevant objects or actions first manifest or disappear. We further identified that such adaptive sensing can be extended to the processing pipeline. More precisely, we execute the compute-heavy transformer encoder-decoder blocks only on frames associated with such anchor moments.

However, we need to tackle two main challenges with this proposed approach. First, the identification of such anchor frames should be computationally inexpensive, while still utilizing features from both event stream and language inputs. Second, even if we can determine such anchor moments (frames), data from an event camera and a traditional camera use two different representations. Hence, we need to devise a unified single-stage STVG model architecture that

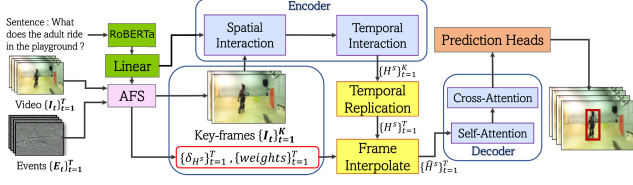


Figure 1. Overall architecture of STVG pipeline of *NeuroViG*

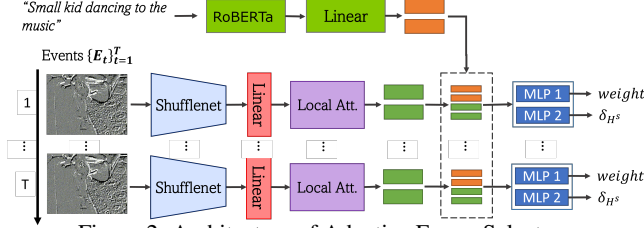


Figure 2. Architecture of Adaptive Frame Selector

accommodates both data representations. Inherently, output from an event camera represents scene changes across frames. Thus, we assume that the processing block from the event camera represents a *differential feature* representation capturing scene changes with respect to the previous frame and the last identified anchor frame. A full-blown STVG pipeline is only executed over the identified anchor frames, while the rest of the redundant frames are approximated with this computed delta representation. To identify these anchor frames and delta feature representations, we propose the **Adaptive Frame Selector** (AFS).

4.1. Adaptive Frame Selector

As shown in Figure 2, AFS takes a series of event frames and language features as input. The raw output of the event camera (Prophesee V2 [4] in our prototype implementation) is described by a stream of $\{x, y, p, t\}$ where $p \in \{0, 1\}$ describes the polarity of the event in either negative or positive direction. We convert this stream-based event representation into a frame representation of shape $[1 \times 448 \times 448]$. This is achieved by aggregating all the events received in a time frame of 33ms resulting in a fixed 30 FPS event stream that is input to the AFS model. This frame representation is passed through a shufflenet network for visual encoding and converted to a 2-dimensional feature vector of size $\mathcal{R}^{T \times D}$, using a fully connected layer of output dimension of $D = 256$. These feature vectors are then passed through a window-based local attention network [5] (window size=10) to model the temporal dependencies in the local neighbourhood to obtain $V_t \in \mathcal{R}^{T \times D}$. Each visual feature V_t is concatenated with the language features computed using a RoBERTa model. Finally, AFS uses 2 separate inexpensive Multi-Layer Perceptron (MLP) models (designed to ensure low-latency processing) with 3 layers and dropout for regularization (dropout probability=0.5). One of these MLPs is used to predict whether a given frame is an anchor-frame or not. The other MLP is used to obtain a feature

representation of the visual differences between the current frame with respect to the nearest anchor-frame. We call this output the *delta feature* representation. These delta features will then be used to interpolate the encoder output to obtain cross-modal attention-based features for non-anchor frames, so that they can be used by the decoder and the prediction heads of the STVG pipeline to predict the spatiotemporal tubes. The two MLPs can be expressed as below.

$$weights = \text{Gumbel}(\text{MLP}[F_l, E_t^f]) \quad \text{where } weights \in \mathcal{R}^{T \times 2} \quad (1)$$

$$\delta_{H^s} = \text{MLP}[F_l, E_t^f] \quad \text{where } \delta_{H^s} \in \mathcal{R}^{T \times D}$$

In the above equation, F_l refers to the language features taken from the same text encoder as STCAT, E_t^f refers to the visual features obtained after computing local attention for an event stream in frame representation of length T frames, while $weights$ determine whether the respective frame is an *anchor-frame* or a *redundant-frame*. We use Gumbel-Softmax activation function, which returns discrete 0 or 1 value for elements in $weights$. When $weights_i[0] = 1$ and $weights_i[1] = 0$, i^{th} frame is deemed to be a *anchor-frame*. When $weights_i[0] = 0$ and $weights_i[1] = 1$, i^{th} frame is deemed to be a *redundant-frame*. We also assume that the first frame of the video is always an anchor-frame (Thus, RGB camera is always activated at the start once). The output of the second MLP, denoted by δ_{H^s} , represents the delta features.

4.2. Loss Function

We propose a modification to the original loss function used in STVG models to reward the model for skipping more RGB frames. At the training stage, we define a *frame-skip* parameter which is a hyper-parameter that determines the percentage of RGB frames that are considered to be redundant in the video. A value closer to 1 would encourage aggressive skipping behaviour, thereby reducing computation and also reducing RGB sensing overhead. A value closer to 0 would encourage conservative skipping behaviour, resulting in higher computational load and sensing overhead. Let this frame-skip parameter be denoted as S and the original loss function of VG be $\mathcal{L}_{VG}$. We formulate the following reward loss in our proposed *NeuroViG* system.

$$\mathcal{L} = \mathcal{L}_{VG} + \lambda_{dIoU} \mathcal{L}_{dIoU}(\hat{b}, b) + \frac{\lambda_w}{N} \sum_{i=0}^N |weights[0]_i - (1-S)| \quad (2)$$

Let us assume that t_s and t_e are the ground truth start and end times of the target object. Then, $b \in [0, 1]^{4 \times (t_e - t_s + 1)}$ and $\hat{b} \in [0, 1]^{4 \times (t_e - t_s + 1)}$ denote the ground truth and predicted bounding box coordinates respectively for the duration where the target object appears. In addition to the original STVG loss function, the second and third terms in

the loss function act as our reward function. The second term rewards the model for correct bounding box prediction using Distance IoU metric [28]. The third term, which is a Mean Absolute Error term rewards the model for skipping a desired fraction of RGB frames as redundant frames. Since S refers to the desired average fraction of frames to be skipped, $1 - S$ refers to the desired average fraction of anchor frames to be identified. λ_{dIoU} and λ_w are two hyper-parameters set for specifying the relative weights for the respective loss terms. These hyper-parameters are set to 1.0 and 2.0 respectively in the training stage.

Figures 1 summarizes this skipping paradigm used in *NeuroViG*, applied to a typical encoder & decoder based STVG pipeline that we shall now describe. The core principle is to activate the traditional RGB camera only for the *anchor-frames* and send them through an expensive STVG pipeline, while the input from an event camera is utilized, together with a late interpolation approach (explained in Equation 3), to obtain the corresponding feature representation for the redundant frames.

4.3. NeuroViG STVG pipeline

We employ the AFS framework to introduce an optimized STVG model within the *NeuroViG* system, which is a derivative of STCAT. Figure 1 depicts the STVG model architecture. Distinguishing itself from existing STVG methods, *NeuroViG* stands out as the first STVG model to accept both event and RGB streams as inputs. This unique feature empowers efficient and rapid processing, even on pervasive devices. The *NeuroViG* model consists of several key components, in addition to the previously explained AFS.

- **Language Encoder:** *NeuroViG* uses ROBERTa [13] as the language encoder. The last hidden state output of ROBERTa is followed by a Linear layer to obtain $F_l \in \mathcal{R}^{L \times d}$ language features where L is the word length and $d = 256$ is the hidden dimension. These language features have already been computed for the AFS stage and can thus simply be reused here.
- **Visual & Multi-modal Encoder:** After the AFS step, we get a series of K anchor-frames, $\{I_t\}_{t=1}^K$. These K frames are passed to the ResNet101 [7] backbone to obtain $F_v \in \mathcal{R}^{K \times HW \times d}$, where H and W are the resolution of the feature maps from the ResNet backbone. Each $\{f_v\}_{i=1}^M$ is concatenated with F_l to obtain the multi-modal feature representation, which is then passed to 2 transformer encoders with 6 layers (similar to STCAT) comprising the spatial interaction and temporal interaction layers. The key idea in the spatial interaction layer is to model spatial relationships in individual frames of the video, whereas the temporal interaction layer models the temporal dependencies across the video.

- **Redundant Frame Interpolation:** As mentioned earlier, we always assume that the first frame of a video is an anchor-frame. While an anchor frame is passed through a complex ResNet visual backbone and an encoder pair to obtain attention based features, the subsequent redundant frames by-pass this complex ResNet based visual encoder and the transformer-based encoder network. In order to obtain the attention based features for such redundant frames, we adopt an interpolation based approach. First, to match the original sequence length of T video frames, we temporally replicate H_i^s till the next $(i + 1)^{\text{th}}$ anchor frame is found. We repeat this for $i = 0$ to $i = K - 1$ till we match the same sequence length of T . Then, for each of the temporally replicated frame H^s , we add (as shown in Equation 3) the corresponding delta features δ_{H^s} calculated during the AFS step to obtain $\hat{H}^s$.

$$\hat{H}^s = H^s + weights[1] * \delta_{H^s} \quad (3)$$

Intuitively, each of these interpolated frames approximates the attention based features of the corresponding redundant frame by combining the attention based features of the previous anchor-frame with the delta features that represent the visual differences of the redundant frame with respect to the anchor frame.

- **Multi-modal Decoder:** After redundant frame interpolation, STCAT proposes a query-guided decoding network, with dual-decoder architecture, to model the temporal interactions among a set of video-text features. We adopt this dual-decoder network as the multi-modal decoder which takes the output from the visual & multi-modal encoder. Multi-modal decoder outputs a feature representation $H^D \in \mathcal{R}^{K \times d}$.
- **Prediction Heads:** We use two MLPs as prediction heads that take H^D as an input: (a) a 3-layer MLP for predicting bounding box coordinates, $\hat{b} \in [0, 1]^{T \times 4}$ and (b) a 2-layer MLP for predicting start and end time probabilities $\hat{\mathcal{T}}_s \in [0, 1]^T$ and $\hat{\mathcal{T}}_e \in [0, 1]^T$. During inference, the start and end times of the target object $\hat{t}_s$ and $\hat{t}_e$ are computed by finding the maximum of joint start and end time probabilities, with invalid combinations (i.e., tuples where $\hat{t}_s \geq \hat{t}_e$) masked out.

NeuroViG is trained for a single epoch with the initial weights taken from the trained STCAT model and the loss function as explained in Equation 2, where $\mathcal{L}_{VG}$ is replaced with the original loss function of STCAT and frame-skip parameter S is set to 0.7.

4.4. NeuroViG System Implementation

Figure 3 shows our proposed system. We use Prophesee V2 [4] as the event camera and Kinect DK [1] as RGB camera and microphone. Both these devices are connected

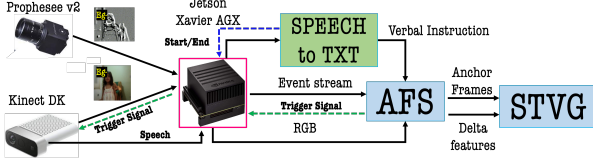


Figure 3. *NeuroViG* System Architecture

to a Jetson Xavier AGX device [3], which executes both the AFS preprocessing component and the rest of the STVG pipeline. Both these devices are not officially supported for integrating with ARM-based devices. Thus, for Prophesee V2, we use their open-sourced SDK, OpenEB, instead of the official MetaVision SDK. For the Kinect DK, we use a cross-compiled version of their SDK for the ARM-based devices. AFS and the rest of the STVG pipeline is implemented using PyTorch. The microphone of the Kinect DK remains continuously active, and we employ a real-time speech-to-text engine [2], running on the AGX device, to convert speech into textual commands. The *NeuroViG* system is initiated upon the user’s command, typically with the phrase ‘start’, and is subsequently terminated upon receiving the user’s ‘done’ command. This action triggers the activation of the event camera. We assume the initial frame is always designated as an anchor-frame, leading to the activation of the RGB camera for a single frame alongside the event camera. AFS then takes the output from the event camera and textual query and sends a trigger signal to the Kinect DK camera once it infers an anchor moment in the video. Such intermittent triggering is also associated with a startup delay of 120ms. However, as current STVG pipelines are usually executed at 5 FPS, we can safely assume that the startup delay of 120ms is tolerable and allows STVG to be executed at 5 FPS without missing frames.

5. Results

We now experimentally evaluate *NeuroViG* vs. alternative state-of-the-art VG models. In general, we follow the same evaluation methods utilized in prior work [9, 24]. In addition to the standard metrics, we shall also consider latency, frames per second (FPS), sensing energy, processing and memory consumption as additional systems-level performance metrics on a Jetson AGX device.

5.1. Datasets & Evaluation Metrics

Datasets: We utilize the VidSTG [27] and HC-STVG [19] benchmark datasets that have also been used in prior work for STVG evaluation. Both datasets are annotated with spatiotemporal tubes with start and end frames, as well as bounding boxes for each frame, corresponding to text queries. **VidSTG** consists of 99,943 sentence descriptions with 44,808 declarative sentences and 55,135 interrogative sentences describing 79 types of objects appearing

in 10,303 different videos. **HC-STVG** is a human-centric STVG dataset with 5,660 videos in multi-person scenes, each associated with one language expression that is related to human attributes or actions. All of the existing STVG datasets are traditional video-based. To obtain the event-based counterpart, we use a V2E simulator [8], which synthesizes realistic event camera data from any real (or synthetic) conventional frame-based video. Thus, we use V2E to convert the existing videos in VidSTG and HC-STVG to the event representation.

Evaluation Metrics: We follow [27] and use m_vIoU , m_tIoU and $vIoU@R$ as accuracy metrics. $vIoU = \frac{1}{|S_u|} \sum_{t \in S_i} IoU(\hat{b}_t, b_t)$, where S_u and S_i refer to the intersection and union between the predicted tube and ground truth tube. $IoU(\hat{b}_t, b_t)$ refers to the IoU score between the predicted bounding box and target bounding box at the frame t . Thus, m_vIoU refers to the mean $vIoU$ score across the dataset and $vIoU@R$ refers to the percentage number of predictions achieving a $vIoU$ score of at least R . $tIoU$ measures the temporal overlap between the predicted and ground truth tubes, defined as $tIoU = \frac{|S_i|}{|S_u|}$. m_tIoU then refers to the mean $tIoU$ across the dataset. We also measure the system-level metrics (inference latency, energy) on the Jetson AGX, which serves as the representative edge device in our *NeuroViG* prototype.

5.2. Characterizing *NeuroViG* Performance with Frame-Skip Parameter

During the training process of our proposed AFS and *NeuroViG*, we introduced a reward loss mechanism to incentivize the model to skip traditional RGB frames, aiming to reduce sensing and processing overhead. In this study, we assess the performance of the *NeuroViG* system while varying the frame-skip parameter. Our goal is to explore the balance between accuracy and energy efficiency.

Initial measurements indicated that the Prophesee V2 and Kinect DK cameras draw an average power of 0.31W and 1.125W, respectively. It was evident that utilizing an event camera should ideally reduce sensing power consumption by approximately 4 times.

To evaluate this, we first examined how the accuracy of the *NeuroViG* model on the VidSTG dataset fluctuates with different frame-skip parameters, as illustrated in Figure 4. Subsequently, we assessed how the sensing energy per frame varies with the frame-skip parameter, as shown in Figure 5. However, it’s important to note that sensing energy is just one component contributing to the overall energy overhead of the *NeuroViG* system.

We also evaluated the processing energy overhead of the system for different values of the frame-skip parameter, as presented in Figure 6. Our findings indicate that increasing the frame-skip parameter reduces both sensing and processing energy overhead but comes at the cost of lower accu-

racy. Conversely, setting the frame-skip parameter below 0.3 results in higher sensing energy consumption compared to a baseline system using only a traditional RGB camera. Based on this comprehensive study, we opted to select a frame-skip parameter of 0.7. This choice strikes a balance between energy efficiency and accuracy, aligning with our goal to optimize the energy-accuracy trade-off

5.3. Performance on Benchmark Datasets

Tables 1 and 2 then summarize the performance of *NeuroViG* (with frame-skip=0.7) against benchmark STVG models on VidSTG and HC-STVG datasets respectively. From our experiments, we observe that the AFS mechanism allows *NeuroViG* to identify only $\sim 35\%$ of the video frames as anchor frames, enabling the remaining frames to bypass the encoder-decoder blocks. However, this results in a minor drop in their accuracy when compared to STCAT. The m_vIoU of *NeuroViG* is just 1.08% shy of STCAT on the human-centric HC-STVG dataset. On VidSTG dataset, *NeuroViG* achieves an m_vIoU that is just about 1.1% short of STCAT for the interrogative sentences and about 0.99% short of STCAT for the declarative sentences. When compared to the TubeDETR, *NeuroViG* is able to achieve higher m_vIoU on all the benchmark tasks and also achieve multi-fold improvement in grounding speed. Overall, we believe that this drop in accuracy is tolerable in light of the fact that nearly 65% of the frames by-passed the RESNET backbone and transformer encoder, and were approximated via an event camera representation instead of a conventional RGB input. We believe that this will translate into significant savings in latency, energy and memory when dealing with longer untrimmed videos in practical use-cases.

We then measured the average FPS, sensing and processing energy per frame on Jetson AGX and compared them against STCAT in Table 3. We found that *NeuroViG* runs 3.82x faster, consumes 2.2x less sensing energy and 4.25x less processing energy. Overall, this results in *NeuroViG* being **4.08x** energy efficient than STCAT, the current state-of-the-art STVG approach. Figure 7 plots the latency of STCAT vs. *NeuroViG*. We see that while the latency for both models increases with the number of frames, However, the rate of increase of latency for *NeuroViG* is significantly lower than that of RGB-based STCAT.

In Table 3, we compared the performance of STVG using existing video processing techniques instead of our proposed *NeuroViG*'s AFS paradigm. STCAT-SparseSampler incorporates sparse sampling as proposed in ClipBERT [11], where a random subset of frames is selected as anchor frames and applied to STCAT. STCAT-MGSampler, on the other hand, adopts the motion-guided frame selection method from MGSampler [29], which selects anchor frames to cover key segments with high motion salience. However, the results show that while these techniques achieve simi-

lar FPS and energy consumption to *NeuroViG*, they lead to a substantial reduction in m_vIoU. While ClipBERT's random sparse sampling may be effective for tasks such as question answering, and MGSampler's motion-based sampling performs well in video action classification, these methods are insufficient for STVG, which requires continuous tracking of target object/region across frames.

5.4. Qualitative analysis: *NeuroViG* skipping

We further analyzed some of the instructions in the dataset to understand how the AFS module picks the anchor-frames. Our analysis revealed that it picks up anchor frames whenever (a) there are static scenes in the video for longer periods (event sensor not generating enough events), (b) visual content partially matches the words in the text, and (c) target object appears in a frame. Observation (a) makes intuitive sense as event cameras inherently fail to generate events when the scene is completely unchanged or static; accordingly, AFS triggers the traditional RGB camera in such scenarios.

Figure 8 shows two such example scenarios from the VidSTG dataset. This figure shows a series of RGB and event frames that are identified as anchor-frames, along with the instruction. In both examples, it appears that the AFS predicts an anchor-frame and subsequently triggers the RGB camera when a keyword or the target object appears in the scene. Furthermore, in the first example, anchor frames are also flagged for static scenes, where the event camera fails to generate sufficient discriminatory events for the STVG pipeline.

5.5. Other variations of *NeuroViG*

We consider additional variations of *NeuroViG* to evaluate the efficacy of our hybrid sensing mechanism. Based on Table 4, it is evident that the performance of *NeuroViG* is significantly degraded when selecting anchor-frames at random (*NeuroViG*-Random). This suggests that the careful selection of anchor-frames is paramount to the performance of *NeuroViG* system. With *NeuroViG*-Vision, we seek to evaluate whether just using event visual feature on AFS is enough to identify anchor-frames. Just using visual features limits the capability of AFS to make an early decision about when the target object appears and only allows it to perceive changes in the scene. As shown in Table 4, while the performance does not degrade as much as in *NeuroViG*-Random, we observe a significant reduction in the accuracy metrics. This demonstrates the importance of using both language and visual features in AFS to predict the anchor-frames in the video. Given the advantages of event cameras, a plausible alternative approach might be to utilize only the DVS-based event camera as input (Event-STCAT), instead of *NeuroViG*'s hybrid sensing mechanism that uses both event & RGB cameras. Such a system would reduce

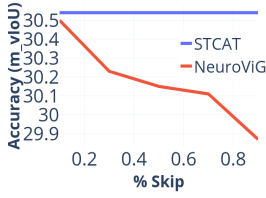


Figure 4. Accuracy (m_vIoU) vs. frame-skip

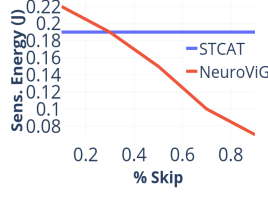


Figure 5. Sensing energy per frame vs. frame-skip

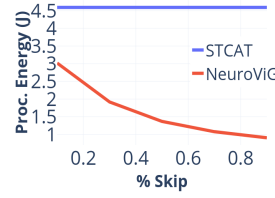


Figure 6. Processing energy/frame vs frame-skip

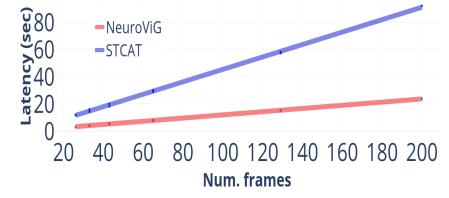


Figure 7. Latency vs num. frames for *NeuroViG* against STCAT

Table 1. Performance comparisons of the state-of-the-art on the VidSTG test set (%)

Methods	Declarative Sentences					Interrogative Sentences				
	m_tIoU	m_vIoU	vIoU@0.3	vIoU@0.5	% frames skipped	m_tIoU	m_vIoU	vIoU@0.3	vIoU@0.5	% frames skipped
Two-Stage:										
STGVT [19]		21.62	29.80	18.94	0	-	-	-	-	0
One-Stage:										
STVGBert [18]		23.97	30.91	18.39	0		22.51	25.97	15.95	0
TubeDETR [24]	48.10	30.40	42.50	28.20	0	46.90	25.70	35.70	23.20	0
STCAT [9]	50.82	33.14	46.20	32.58	0	49.67	28.22	39.24	26.63	0
<i>NeuroViG</i> (Ours)	49.01	32.15	44.89	31.06	66	48.89	27.12	38.09	25.43	64

Table 2. Performance on the HC-STVG test set (%)

Methods	m_tIoU	m_vIoU	vIoU @0.3	vIoU @0.5	% frames skipped
Two-Stage:					
STVGT	-	18.15	26.81	9.48	0
One-Stage:					
STVGBert	-	20.42	29.37	11.31	0
TubeDETR	43.70	32.40	49.80	23.50	0
STCAT	49.44	35.09	57.67	30.09	0
<i>NeuroViG</i>	48.65	34.01	56.12	28.91	65

Table 3. Accuracy vs FPS vs Energy trade-off for *NeuroViG* against other baseline STVG systems

Model	m_vIoU	FPS	Energy per frame (J)		
			Sensing	Processing	Total
STCAT	30.54	2.2	0.22	4.59	4.81
STCAT-SparseSampler	12.19	8.8	0.22	1.04	1.26
STCAT-MGSampler	19.11	8.6	0.22	1.02	1.24
<i>NeuroViG</i>	30.11	8.4	0.10	1.08	1.18

Table 4. Ablation study of *NeuroViG*

Methods	m_tIoU	m_vIoU	vIoU@0.3	vIoU@0.5
<i>NeuroViG</i>	48.62	29.40	41.67	28.07
<i>NeuroViG</i> -Random	31.12	16.15	20.21	14.76
<i>NeuroViG</i> -Vision	40.14	22.98	33.54	21.12
STCAT-Event	34.17	19.76	28.16	17.19

6. Conclusion

We present *NeuroViG*, a novel system combining an event camera and an RGB camera for energy-efficient spatio-temporal video grounding. Our Adaptive Frame Selection (AFS) module identifies key anchor frames using event streams and language cues, activating the RGB sensor only when needed. The system’s dynamic pipeline improves efficiency by selectively invoking a transformer-based DNN for these frames. *NeuroViG* is the first to use a neuromorphic event camera for STVG, achieving a 4x energy reduction and 3.8x faster processing with only 1% accuracy loss.

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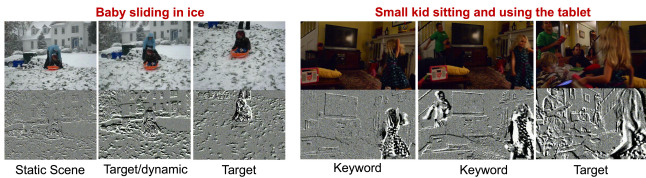


Figure 8. Qualitative analysis of the functionality of AFS

the sensing overhead $\sim 4x$. However, we observe a similar significant degradation in accuracy (m_vIoU drop of 40+%) when using Event-STCAT. This suggests that, while event cameras do provide a good energy-aware approximation to traditional RGB camera sensing, a combination of event+RGB features is paramount for STVG.

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